

Credit Card Fraud Detection Using a Statistical Model

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1. Introduction – The Anatomy of Credit Card Fraud

1.1. The Business and Accounting Significance of the Problem

With the explosive growth of digital transactions, credit card fraud has evolved into a global industry. For a financial institution, fraud represents not only direct accounting losses (written-off receivables, compensations) but also something even more severe: the erosion of trust capital. Traditional, rule-based internal control and risk management systems have now reached their limits. The manual inspection of thousands of transactions arriving per second is impossible, while rigid limit systems either generate too many false alarms or remain highly vulnerable.

According to industry analyses by the Nilson Report, the global damage resulting from credit card fraud amounts to tens of billions of dollars annually, and this figure is continuously growing with the dominance of e-commerce. (Globe Newswire, 2025)

Financial institutions have traditionally applied static rule-based systems for defense, such as geographical restrictions (for example, if a Hungarian card is suddenly used in Indonesia) or amount limits. (Palanisamy, 2025)

However, these systems are extremely rigid, and modern cybercriminals can easily bypass them by masking transactions to resemble normal customer behavior.

In this challenging environment, IT and finance intersect. Machine learning and predictive analytics enable banks to proactively filter out suspicious patterns in real time, minimizing financial loss while maintaining the customer experience. This essay documents such an analytical process, from the processing of raw data to executive decision support.

1.2. Presentation of the Examined Dataset

For the analysis, I used an open database from the Kaggle platform containing the transactions of European cardholders, known as the Credit Card Fraud Detection dataset. The dataset records two days (48 hours) of banking traffic, totaling 284,807 transactions.

Due to bank secrecy and GDPR regulations, the customers' personal data and card details are not included in the database. The dataset contains 28 so-called Principal Component columns

(numeric variables ranging from V1 to V28). These are the results of a Principal Component Analysis (PCA) mathematical transformation, which preserve the statistical properties of the original data but cannot be reverse-engineered. Only two variables remained in their original form: Time (the number of seconds elapsed between transactions, starting from the first transaction) and Amount (the transaction amount in euros). Additionally, there is a binary indicator, where 0 represents a legitimate transaction and 1 denotes a proven fraudulent transaction.

1.3. Research Objective and Methodology of the Analysis

The primary objective of the analysis is to build a statistical classification model—specifically, Logistic Regression—using the SAS OnDemand for Academics software to detect fraud. The greatest methodological challenge of the project is the extreme imbalance of the data. The basic statistics run during the examination of the binary indicator numerically confirmed the severity of the problem and the difficulty of the task, revealing that out of nearly 285,000 transactions, only 492 cases (0.17%) were actual fraud.

The essay subsequently presents the process of data cleaning and scaling, the sampling techniques (Under-sampling), the construction of the predictive model, and finally, the accounting and business interpretation of the resulting confusion matrix.

The analysis of average transaction amounts also revealed a telling pattern. While the average value of normal purchases was relatively lower, fraudsters on average attempted slightly higher amounts, yet they carefully avoided extreme outliers of tens of thousands of euros. This behavior is presumably deliberate: criminals try to stay under the radar to avoid immediate, automatic limit-blocking by banking systems.

2. Requirement Specification and IT Solution

The primary objective of the research and the IT solution to be developed is the creation of a real-time, predictive decision support system capable of the automatic, second-by-second classification of transactions. The fundamental business and IT requirement for the system is that the mathematical model filters out actual fraud with the highest possible accuracy while keeping the number of false alarms to a minimum. Keeping the proportion of falsely accused honest customers—who are thereby prevented from making payments—low is a top business priority, as an overly strictly calibrated system carries severe customer dissatisfaction and reputational risks for the bank. Accordingly, the quality and success of the model are determined not by general statistical accuracy, but by the delicate balance between the proportion of caught fraudsters and precision (the rate of false alarms). Furthermore, during the IT implementation, it is advisable to use the SAS software environment, which, with its robust data management capabilities, can handle large-scale datasets meeting industry standards.

Although Microsoft Excel is theoretically capable of handling a row count of this size, in practice, the simultaneous, complex statistical calculation, data scaling, and predictive modeling of nearly 285,000 records and their 31 variables would generate such a high memory and processor demand that it would result in significant processing time and system stability risks. In contrast, the robust, professional architecture and code-based processing logic of SAS allow for the efficient, secure, and fast analysis of large-scale datasets that meet industry standards. Moreover, the SAS OnDemand for Academics version is a cloud-based solution, which means that running computationally intensive algorithms does not burden the local computer but utilizes the high-performance servers of the SAS Institute. This architecture makes the platform particularly ideal for research and academic essay writing purposes, as it enables complex Big Data analyses with essentially zero local hardware requirements.

3. Feasibility Study

3.1. The Problem of Statistical Noise

To successfully solve the task, it is essential to select the appropriate data preparation and modeling methodologies, as the examined dataset represents a classic statistical problem known as rare events. Traditional machine learning algorithms typically perform extremely poorly in cases where an extreme imbalance can be observed between the classes of the target variable. (Krawczyk, 2016)

If the training of the logistic regression model were to be conducted directly on the original dataset containing ninety-nine percent legitimate transactions, the algorithm would merely learn the characteristics of the majority class, while treating the rare but critical fraud attempts as simple statistical noise and ignoring them. The literature refers to this phenomenon as the accuracy paradox. Since the proportion of fraud in the database is only 0.17 percent, a naive algorithm that predicts every single transaction to be legitimate would achieve an overall accuracy of 99.83 percent. Although this metric looks excellent on paper, the business value of the model would be zero, as it would fail to find the very crimes it is supposed to filter out.

With such extremely asymmetric datasets, traditional models optimize exclusively for the majority class, thus mere statistical accuracy as an evaluation metric is highly misleading. For this reason, the primary condition for feasibility is the preliminary, artificial transformation of the data. (Chawla et al., 2004)

3.2. Solution to the Noise: Under-sampling

To eliminate this problem, the application of the under-sampling technique during the data management phase of the analysis is the most effective solution. The essence of this step is to select, using a random number generator, exactly as many observations from the majority, legitimate transactions as there are in the minority class, that is, among the frauds. With this methodological step, an artificially balanced training dataset with a fifty-fifty percent ratio is created. This new, balanced structure enables the predictive algorithm to focus on identifying the hidden dimensions and patterns characteristic of fraudsters, without the vast amount of legitimate data suppressing these subtle signals. (Batista et al., 2004.)

The literature frequently refers to an alternative solution, the Synthetic Minority Over-sampling Technique (SMOTE), which does not discard legitimate transactions but rather generates artificial new data points from the frauds. (Chawla et al., 2002)

In the case of the current research, however, the application of under-sampling was more justified from both a technological and resource management perspective. The initial database contains nearly three hundred thousand rows; over-sampling would have drastically increased this dataset to well over half a million records, which would have required significant and unnecessary processing time in the SAS software. The more compact, noise-free training dataset obtained through under-sampling allowed for more efficient execution and cleaner model building.

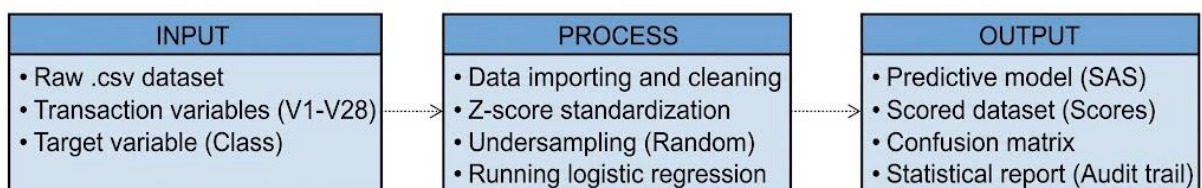
3.3. Standardizing Variables to Eliminate Biases

Prior to the sampling procedure, another critical data cleaning step—the scaling of variables—is also essential for feasibility. Since the values of the Amount variable, which contains the original monetary sums, differ by orders of magnitude from the low, decimal values of the encrypted, previously transformed principal components (V1-V28), the amount column must be standardized. Without this transformation, the algorithm's computational mechanism would incorrectly assign greater weight to the variable with a higher nominal value (larger numbers). If the variables are not on a common scale, training the model will take an unreasonably long time, and the final prediction will be statistically biased. Standardization ensures that the mathematical model can examine every dimension with equal opportunity and identical weight during the optimization process. (Géron, 2019.)

4. Logical and Physical System Design

4.1. Input – Process – Output (IPO) diagram

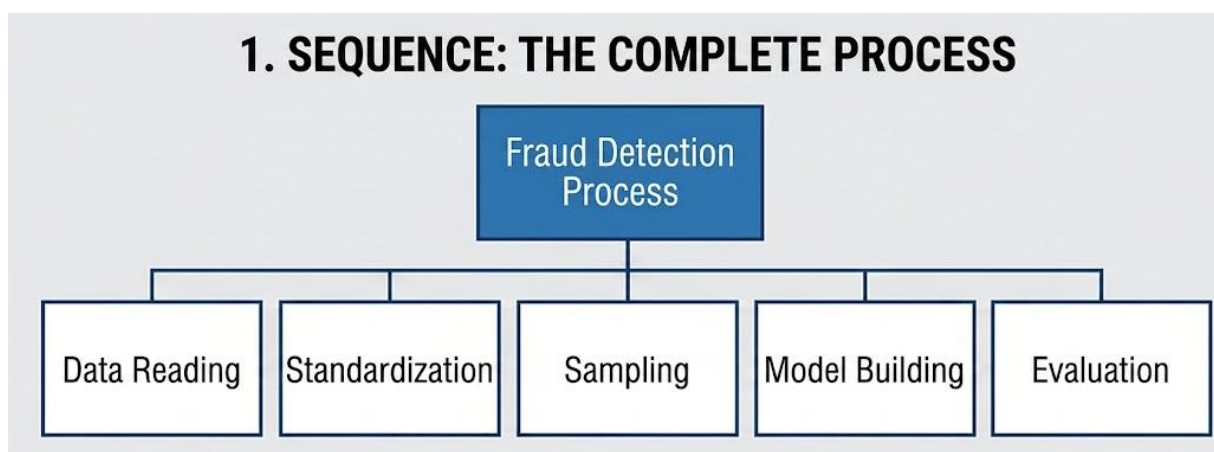
The IPO diagram presents the fundamental data flow framework of the system, separating resources from operations. The input layer contains the raw .csv file consisting of 284,807 records and the transaction variables. In the processing phase, the SAS software performs data cleaning, Z-score standardization, and logistic regression analysis. Finally, the output layer provides the classified transactions, probability scores, and the statistical confusion matrix for decision-makers.



4.2. Jackson Diagrams

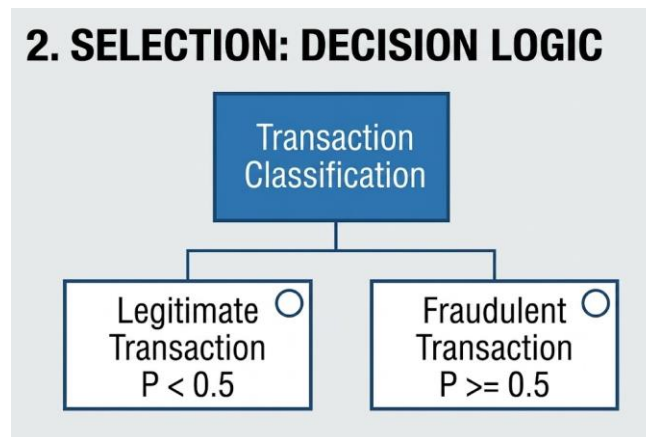
4.2.1. Sequence – The Complete Process

The sequence diagram illustrates the linear, interdependent phases of the software implementation. The process follows a strict order: from data ingestion, through the standardization of variables and sampling, leading up to the training of the model.



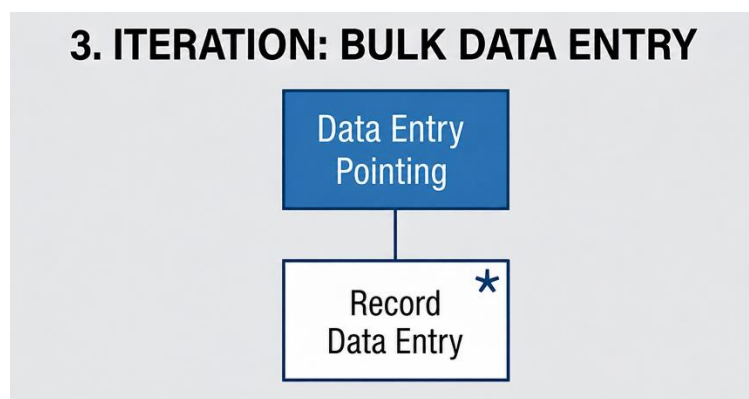
4.2.2. Selection – Decision Logic

The selection diagram models the binary decision logic of the model. During the evaluation of each individual transaction, the system arrives at a logical decision point: based on the probability calculated by the predictive algorithm, the transaction is classified into either the 'legitimate' or the 'fraudulent' class.



4.2.3. Iteration – Mass Processing

The iteration diagram presents the system's cyclic algorithm for processing large volumes of data. Since the banking dataset contains hundreds of thousands of records, the SAS software performs the exact same evaluation operation on every single row (transaction) within an iterative loop. This repeating structure guarantees comprehensive verification, ensuring that not a single transaction remains unchecked in the entire database.



5. Implementation and Applied Algorithms

During the practical implementation, the first step was importing the raw, semicolon-delimited, text-format .csv dataset into the cloud-based environment of SAS OnDemand for Academics. In the .csv file, each row represents a record, and each semicolon represents an attribute. I carried out the import using SAS's built-in importing procedure. The system automatically recognized the names and types of the variables, and then created the initial table ready for analysis in the server's working directory.

```
%web_drop_table(WORK.IMPORT);
```

```
FILENAME REFFILE '/home/u64472364/Card Fraud Essay/creditcard.csv';
```

```
PROC IMPORT DATAFILE=REFFILE
```

```
    DBMS=CSV
```

```
    OUT=WORK.IMPORT;
```

```
    GETNAMES=YES;
```

```
RUN;
```

```
PROC CONTENTS DATA=WORK.IMPORT; RUN;
```

```
%web_open_table(WORK.IMPORT);
```

Following the successful loading of the data and the execution of exploratory statistics, as a prerequisite for model building, I performed the scaling of the variables. As stated in the feasibility study, the *Amount* variable representing the transaction amount differed by orders of magnitude from the encrypted principal components. I bridged this problem by applying a standardization procedure, thereby ensuring that the logistic regression algorithm would not assign unreasonably large weights to the higher nominal values.

```
proc stdize data=WORK.IMPORT method=std out=WORK.FRAUD_SCALED;
```

```
    var Amount;
```

```
run;
```

Following the scaling, as part of a data management step, I split the legitimate and fraudulent transactions into two separate tables based on the target variable, setting the stage for the sampling procedure.

```
data WORK.CSALAS WORK.LEGALIS;
```

```
    set WORK.FRAUD_SCALED;
```

```
    if Class = 1 then output WORK.CSALAS;
```

```
    else output WORK.LEGALIS;
```

```
run;
```

The most critical point of the task was the creation of the artificially balanced training dataset using the under-sampling procedure. To this end, from the table containing more than 280,000 legitimate transactions, I selected exactly 492 observations using the appropriate SAS procedure and the Simple Random Sampling method. I merged this sample with the table containing all registered frauds. This method resulted in a perfectly balanced working dataset consisting of 984 rows, on which the predictive model was able to reliably identify the hidden patterns of abuse without the majority class suppressing them.

```
/* Alul-mintavételezés: 492 legális tranzakció kiválasztása és egyesítése a csalásokkal */  
proc surveyselect data=WORK.LEGALIS method=srs n=492 out=WORK.LEGALIS_SAMPLE;  
run;  
  
data WORK.MODEL_ADAT;  
  set WORK.CSALAS WORK.LEGALIS_SAMPLE;  
run;
```

In the actual machine learning phase, the construction of the logistic regression model followed. During the parameterization of the model, I applied the stepwise variable selection method, which automatically examined all available variables and filtered out the statistically non-significant parameters, thereby preventing model overfitting and increasing the stability of the prediction. Through the appropriate code instruction, the software immediately applied the algorithm trained on the balanced sample to the entire original dataset of nearly three hundred thousand rows, calculating the mathematical probability of fraud occurrence for every single incoming transaction.

```
/* Logisztikus regresszió futtatása és a teljes adathalmaz kiértékelése */  
proc logistic data=WORK.MODEL_ADAT descending;  
  model Class = Amount V1-V28 / selection=stepwise;  
  score data=WORK.FRAUD_SCALED out=WORK.SCORED_DATA;  
run;
```

6. Summary of Results and Business Evaluation

To determine the real business value of the logistic regression model trained on the artificially balanced sample, the system had to be tested on the complete, original dataset containing nearly three hundred thousand transactions. The evaluation of the model was conducted along a fifty percent intervention threshold, meaning that the system automatically blocked and classified as fraud those transactions where the statistical risk exceeded this level. The software aggregated the results of the run in the form of a confusion matrix, which crystal-clearly demonstrated the accounting and customer relationship impacts of the IT solution.

```
/* A kiértékelés - Hány csalót fogtunk meg? */  
data WORK.EVALUATION;  
  set WORK.SCORED_DATA;  
  /* A P_1 oszlop tartalmazza a csalás valószínűségét.  
     Ha ez nagyobb mint 0.5 (50%), akkor a modell riaszt (1). */  
  if P_1 > 0.5 then Predicted_Class = 1;  
  else Predicted_Class = 0;  
run;  
  
title "Üzleti Eredmény: Tényleges vs. Modell által becsült (Konfúziós Mátrix)";  
proc freq data=WORK.EVALUATION;  
  tables Class * Predicted_Class / norow nocol nopercnt;  
run;  
title;
```

The most important proof of the model's success was its ability to identify actual frauds, a metric technically referred to as "sensitivity." Out of the 492 actual, registered abuses in the database, the predictive system successfully identified 448 cases. This represents a 91.1 percent detection success rate, which is considered an excellent result even by industry standards. From an accounting perspective, this ratio translates directly into saved capital and avoided compensation liabilities, as the bank was able to prevent theft in more than ninety-one percent of the cases. Only 44 sophisticated frauds slipped through the system's net (false negative cases), representing the remaining, yet maximally tolerable risk loss for the bank.

The other side of the coin, namely the price of IT security, is shown by the number of false alarms (false positive cases). The model falsely indicated fraud in 8,267 cases for transactions that were completely legitimate in reality. Although this number may seem high in absolute terms, compared to the more than two hundred eighty thousand normal transactions, this error rate is extremely low, significantly sparing customer service capacities. In the credit card market, the risk management strategy of financial institutions consciously accepts this

compromise: it is a much more rational business step to verify a suspicious but legitimate transaction via an automated call or SMS than to let cybercriminals empty a customer's account. The cost of the additional administrative burden stemming from false alarms is dwarfed by the financial and reputational gains derived from prevented frauds.

Business Result: Actual vs. Model Estimated (Confusion Matrix)

The FREQ Procedure

Frequency		Table of Class by Predicted_Class		
Class		Predicted_Class		
		0	1	Total
0		276048	8267	284315
1		44	448	492
Total		276092	8715	284807

In conclusion, the coordinated work of the IT and financial-accounting fields, the appropriate data preparation methodologies (such as under-sampling), and the application of predictive machine learning models are indispensable in modern banking security. The constructed SAS model proved that it is capable of filtering out criminal patterns with high accuracy even from the hidden, encrypted dimensions of transactions. The direction of future developments could be the dynamic fine-tuning of intervention thresholds adapted to customer profiles, whereby the number of false alarms can be further reduced while maintaining the accuracy of fraud detection.

7. References

7.1. Publications:

1. Advanced Rule-Based Fraud Detection Systems In Payment Processing, Arun Palanisamy, 2025.
2. Learning from imbalanced data: open challenges and future directions, Bartosz Krawczyk, 2016.
3. Special Issue on Learning from Imbalanced Data Sets, Chawla et al., 2004.
4. SMOTE: Synthetic Minority Over-sampling Technique, Chawla et al., 2002.
5. A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data, Batista et al., 2004.
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7.2. Online Articles:

1. Payment Card Fraud Losses Approach \$34 Billion, Globe Newswire, 2025.
(<https://www.fintechfutures.com/press-releases/payment-card-fraud-losses-approach-34-billion>)

7.3. Datasets:

1. Kaggle Credit Card Fraud Detection Dataset
(<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud/data>)